

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO2:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr DUMPALA PRASANTH

QUESTIONS: 10

Total Marks: 50

Annexure A
Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. **(5 Marks)**
2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. **(5 Marks)**
3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time. **(5 Marks)**
4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. **(5 Marks)**
5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism. **(5 Marks)**

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. **(5 Marks)**

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization. **(5 Marks)**

8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency. **(5 Marks)**

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance. **(5 Marks)**

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

Code of Conduct:

*I K Uday Kiran Reddy (192425211)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:k Uday

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q1. Reinforcement Learning Framework for Ride-Sharing

Problem Understanding:

A ride-sharing company needs to efficiently match drivers with passengers. The objective is to minimize passenger waiting time, reduce unnecessary driver travel, and improve overall service quality.

RL Framework:

Agent: Ride-sharing matching system.

Environment: City road network containing passengers and available drivers.

State Space:

The state can include:

Locations of available drivers

Locations of passengers

Number of waiting passengers

Traffic conditions

Driver availability

Current estimated waiting time

Action Space:

The agent can:

Assign a driver to a passenger

Keep a driver available

Redirect a driver toward a high-demand area

Reward Function:

A suitable reward can be:

$\text{Reward} = - \text{passenger waiting time} - \text{driver travel distance} - \text{cancellation penalty}$

A positive reward can be given for successful and fast matching.

Policy:

The policy determines which driver should be assigned to which passenger for a given state.

Working:

The system observes the current state, selects a matching action, receives a reward based on waiting time and service quality, and updates its policy.

Conclusion:

The RL system learns matching decisions that reduce waiting time and improve driver utilization and customer satisfaction.

Q2. RL for Personalized E-Learning Recommendations

Problem Understanding:

An e-learning platform wants to recommend study materials according to each student's learning behavior and needs. The challenge is balancing familiar content with new content.

RL Framework:

Agent: Recommendation system.

Environment: E-learning platform and student.

State: Student's learning history, completed lessons, performance, interests, and current learning level.

Actions: Recommend a particular video, lesson, quiz, exercise, or new topic.

Reward: Positive reward for improved performance, completion, engagement, and useful recommendations.

Exploration:

Exploration means recommending new or less-tested study materials to discover whether they are useful to the student.

Exploitation:

Exploitation means recommending materials that have previously produced good results for the student.

Conclusion:

A good balance between exploration and exploitation allows the system to personalize learning while discovering better educational resources.

Q3. RL Framework for a Robotic Vacuum Cleaner

Problem Understanding:

A robotic vacuum cleaner must clean the maximum area efficiently while avoiding obstacles such as furniture, walls, stairs, and other objects.

RL Framework:

Agent: Robotic vacuum cleaner.

Environment: House or room.

State Space: Robot location, cleaned/uncleaned areas, obstacle locations, battery level, and sensor information.

Action Space: Move forward, move backward, turn left, turn right, stop, or return to charging station.

Reward Function:

Positive reward for cleaning a new area.

Negative reward for collision.

Negative reward for unnecessary movement.

Negative reward for excessive battery consumption.

Positive reward for successfully returning to the charging station.

Policy:

The policy maps the current state to the best action:

$\pi(s) \rightarrow a$

For example, if an obstacle is detected in front of the robot, the learned policy may select "turn left."

Role of Value Function:

The value function estimates the expected future reward from a state:

$V(s)$ = expected cumulative future reward

Initially, the robot may make inefficient movements. As it learns from rewards, the value function becomes more accurate and helps the robot select states and actions that lead to better cleaning performance.

Conclusion:

The policy determines the robot's actions, while the value function helps evaluate which states are more beneficial. Continuous learning improves cleaning efficiency and obstacle avoidance.

Q4. RL-Based Cryptocurrency Trading System

Problem Understanding:

A cryptocurrency trading system uses RL to select Buy, Sell, or Hold actions. The major challenge is designing a reward that encourages useful trading behavior rather than undesirable strategies.

RL Framework:

Agent: Trading agent.

Environment: Cryptocurrency market.

State: Current price, historical prices, portfolio balance, holdings, trading volume, and market indicators.

Actions: Buy, Sell, Hold.

Reward: Change in portfolio value after considering trading costs and risk.

A simplified reward can be:

$\text{Reward} = \text{Portfolio Profit} - \text{Transaction Cost} - \text{Risk Penalty}$

Reward Design Challenges:

Short-term vs long-term profit:

A reward based only on immediate profit may cause excessive trading.

Transaction costs:

Ignoring fees can make an apparently profitable strategy unrealistic.

Risk:

The agent may take excessive risk to obtain high rewards.

Market volatility:

Sudden price changes can produce unstable learning.

Effect of Poor Reward Design:

If rewards only consider profit, the agent may learn risky or excessive trading strategies. If rewards are poorly scaled, learning can become unstable.

Conclusion:

A well-designed reward should consider profit, transaction costs, risk, and long-term performance so that the learned policy produces more reliable decisions.

Q5. RL-Based Smart Grid Electricity Distribution

Problem Understanding:

A smart grid must distribute electricity efficiently while maintaining supply-demand balance and minimizing energy wastage and operating costs.

RL Model:

Agent:

Smart-grid control system.

Environment:

Electricity generation, storage, consumers, and distribution network.

State Space:

Current electricity demand

Available generation

Battery/storage level

Electricity price

Renewable energy availability

Grid load

Action Space:

Increase/decrease power allocation

Charge/discharge batteries

Allocate renewable energy

Adjust distribution according to demand

Reward Function:

Reward = – energy wastage – operating cost – supply-demand imbalance

Additional positive reward can be provided when electricity demand is satisfied efficiently.

Working:

The system observes current grid conditions.
The agent selects an electricity-distribution action.
The environment changes according to the action.
The system receives a reward.
The policy/value estimates are updated.
The process continues to improve future decisions.

Conclusion:

The RL system can learn an efficient distribution policy that balances demand, reduces wastage, and improves utilization of available energy.

Q6. RL-Based Healthcare Medication Dosage

Problem Understanding:

The assessment asks for an RL framework in which a healthcare monitoring system adjusts medication dosage based on patient-related states and explains how learning improves outcomes.

RL Framework:

Agent: Healthcare decision-support system.

Environment: Patient monitoring environment.

State Space: Relevant monitored measurements, patient response to previous dosage, treatment history, and other clinically appropriate observations.

Action Space: Select among clinically approved dosage adjustments.

Reward Function: Reward can reflect improvement toward the desired clinical target while penalizing adverse responses or unsafe deviations.

Policy:

The policy maps the observed patient state to an appropriate treatment action:

$\pi(s) \rightarrow a$

Learning Process:

The system observes the patient's current state.

A treatment action is selected.

The patient's response is observed.

A reward is calculated.

The value of the resulting state is updated.

Future decisions are improved based on accumulated experience.

Role of Value Function:

The value function estimates the expected long-term benefit of a state.

This helps the system prefer decisions that lead to better long-term outcomes rather than focusing only on immediate changes.

Conclusion:

RL can provide an adaptive decision-support framework, but medication decisions require strict clinical supervision, safety constraints, and validation rather than unrestricted autonomous learning.

Q7. RL-Based Drone Delivery Navigation

Problem Understanding:

A drone delivery system needs to select efficient routes while responding to obstacles, weather, battery limitations, and changing environmental conditions. The question specifically focuses on how exploration and exploitation influence route optimization.

RL Framework:

Agent: Delivery drone.

Environment: Airspace and delivery area.

State: Drone position, destination, battery level, obstacles, weather conditions, and current route.

Actions: Move in different directions, change altitude, or select an alternative route.

Reward: Positive reward for reaching the destination efficiently; negative reward for excessive distance, delay, energy consumption, or unsafe route choices.

Exploration:

The drone tries alternative routes to discover potentially shorter or more energy-efficient paths.

Exploitation:

The drone selects routes that have already produced good results.

Conclusion:

Exploration helps discover improved routes, while exploitation uses learned knowledge. Balancing both allows the drone to optimize navigation over time.

Q8. RL Framework for a Manufacturing Robot

Problem Understanding:

A manufacturing robot must perform assembly operations efficiently while minimizing errors, unnecessary movements, and production time.

RL Framework:

Agent: Manufacturing robot.

Environment: Assembly workstation.

State Space: Robot position, component position, assembly stage, tool status, and sensor information.

Action Space: Pick a component, move, rotate, place, assemble, inspect, or change tool.

Reward Function:

Positive reward for successful assembly.

Negative reward for errors.

Negative reward for excessive movement.

Negative reward for excessive assembly time.

Positive reward for completing operations accurately and efficiently.

Policy:

The policy determines the action the robot should perform for a particular assembly state.

Value Function:

The value function estimates the expected cumulative reward from each state.

For example, if one assembly state usually leads to successful and fast completion, it will receive a higher value than a state associated with repeated errors.

Learning Process:

State → Policy → Action → Reward → New State → Value Update

Over repeated operations, the robot learns which actions produce higher long-term rewards.

Conclusion:

The policy guides the robot's decisions, while the value function helps evaluate future outcomes. Together they improve assembly speed, accuracy, and efficiency.

Q9. RL-Based Online Advertising System

Problem Understanding:

An online advertising system uses RL to maximize click-through rate (CTR). The reward must be designed carefully so that the system does not simply maximize clicks at the expense of other important objectives.

RL Framework:

Agent: Advertisement recommendation system.

Environment: Online users and advertising platform.

State: User context, previous interactions, advertisement characteristics, time, and browsing behavior.

Actions: Select an advertisement to display.

Reward: Reward can be based on useful user engagement, such as a click, while also considering longer-term objectives.

Reward Design Challenges:

Click quality:

Maximizing clicks alone may encourage irrelevant advertisements.

Short-term vs long-term objectives:

A user may click an advertisement but have a poor overall experience.

User satisfaction:

Excessive or irrelevant advertisements may reduce user satisfaction.

Bias:

Historical interaction data can cause the system to repeatedly favor already popular advertisements.

Impact of Poor Reward Design:

If the reward is only "1 for click and 0 otherwise," the system may learn to favor advertisements that generate clicks without considering relevance or long-term user engagement.

Better Reward Concept:

Reward = useful engagement + long-term satisfaction – negative user experience

Conclusion:

Careful reward design ensures that RL optimizes meaningful advertising performance rather than only increasing immediate clicks.

Q10. RL-Based Smart Irrigation System

Problem Understanding:

A smart irrigation system must provide sufficient water to crops while minimizing unnecessary water consumption. The assessment specifically requires states, actions, and a reward function.

RL Framework:

Agent:

Smart irrigation controller.

Environment:

Agricultural field and surrounding weather conditions.

State Space:

Soil moisture

Temperature

Humidity

Recent rainfall

Weather forecast information

Crop growth stage

Current water availability

Action Space:

Do not irrigate

Irrigate with low amount of water

Irrigate with medium amount

Irrigate with high amount

Reward Function:

Reward = crop water-satisfaction benefit – water consumption penalty

A positive reward is provided when the crop receives an appropriate amount of water. A negative reward is given for excessive water usage or insufficient irrigation.

Working:

Sensors collect soil and environmental data.

The RL agent observes the current state.

The policy selects an irrigation action.

The system observes the resulting soil moisture and crop conditions.

A reward is calculated.
The agent updates its value estimates/policy.
Future irrigation decisions become more efficient.

Conclusion:

The RL framework can learn an adaptive irrigation policy that maintains suitable soil moisture while reducing unnecessary water consumption.